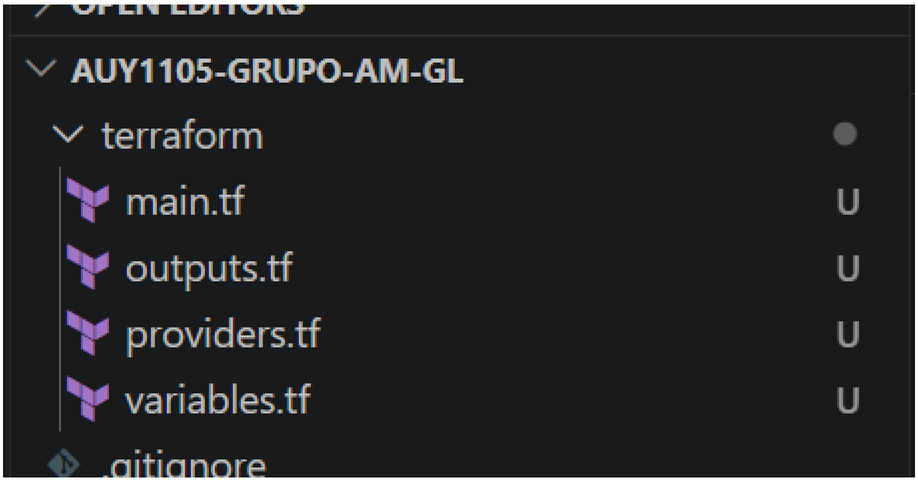

AUY1105-GRUPO-AM-GL
terraform
main.tf U
outputs.tf U
providers.tf U
variables.tf U
.gitignore


providers.tf U
variables.tf U
terraform > providers.tf > provider "aws"
terraform {
required_providers {
aws = {
source = "hashicorp/aws"
version = "~> 5.0"
}
}
required_version = ">= 1.0"
}
provider "aws" {
region = var.aws_region
}

providers.tf U   variables.tf U

terraform > variables.tf > variable "course"

```
variable "aws_region" {
  description = "Región de AWS"
  type        = string
  default     = "us-east-1"
}

variable "project_name" {
  description = "Nombre del proyecto"
  type        = string
  default     = "amgl"
}

variable "course" {
  description = "Sigla del curso"
  type        = string
  default     = "AUY1105"
}
```

providers.tf U | main.tf U | variables.tf U

terraform > main.tf > resource "aws_instance" "main"

```hcl
# VPC
resource "aws_vpc" "main" {
  cidr_block = "10.1.0.0/16"

  tags = {
    Name = "${var.course}-${var.project_name}-vpc"
  }
}

# Subnet pública
resource "aws_subnet" "public" {
  vpc_id            = aws_vpc.main.id
  cidr_block        = "10.1.1.0/24"
  availability_zone = "${var.aws_region}a"

  tags = {
    Name = "${var.course}-${var.project_name}-subnet"
  }
}
```

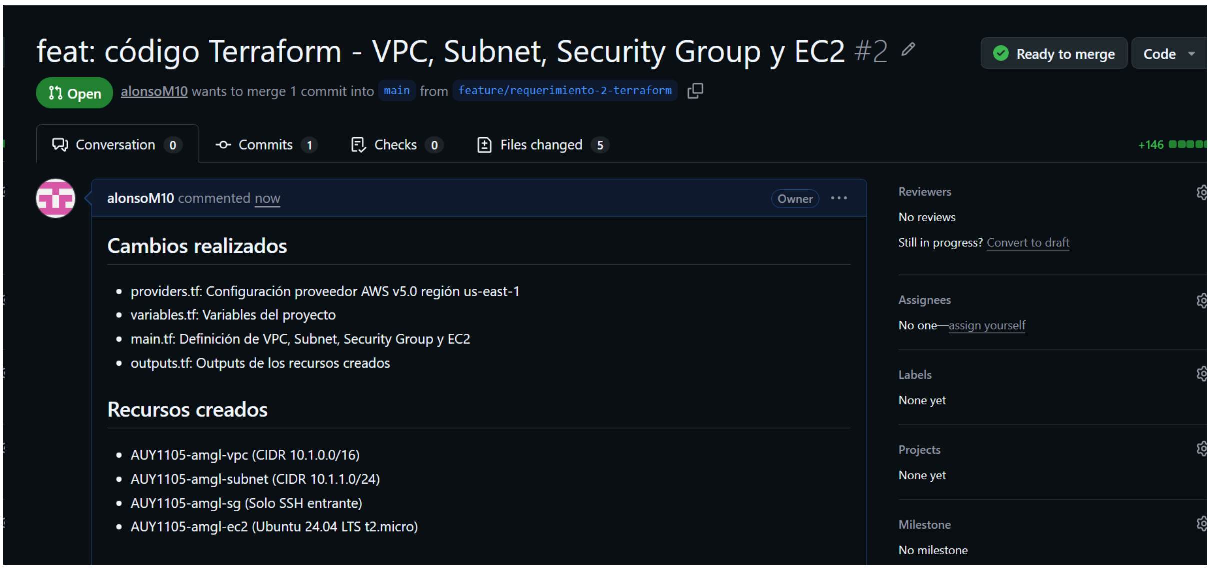
providers.tf U | main.tf U | outputs.tf U | variables.tf U

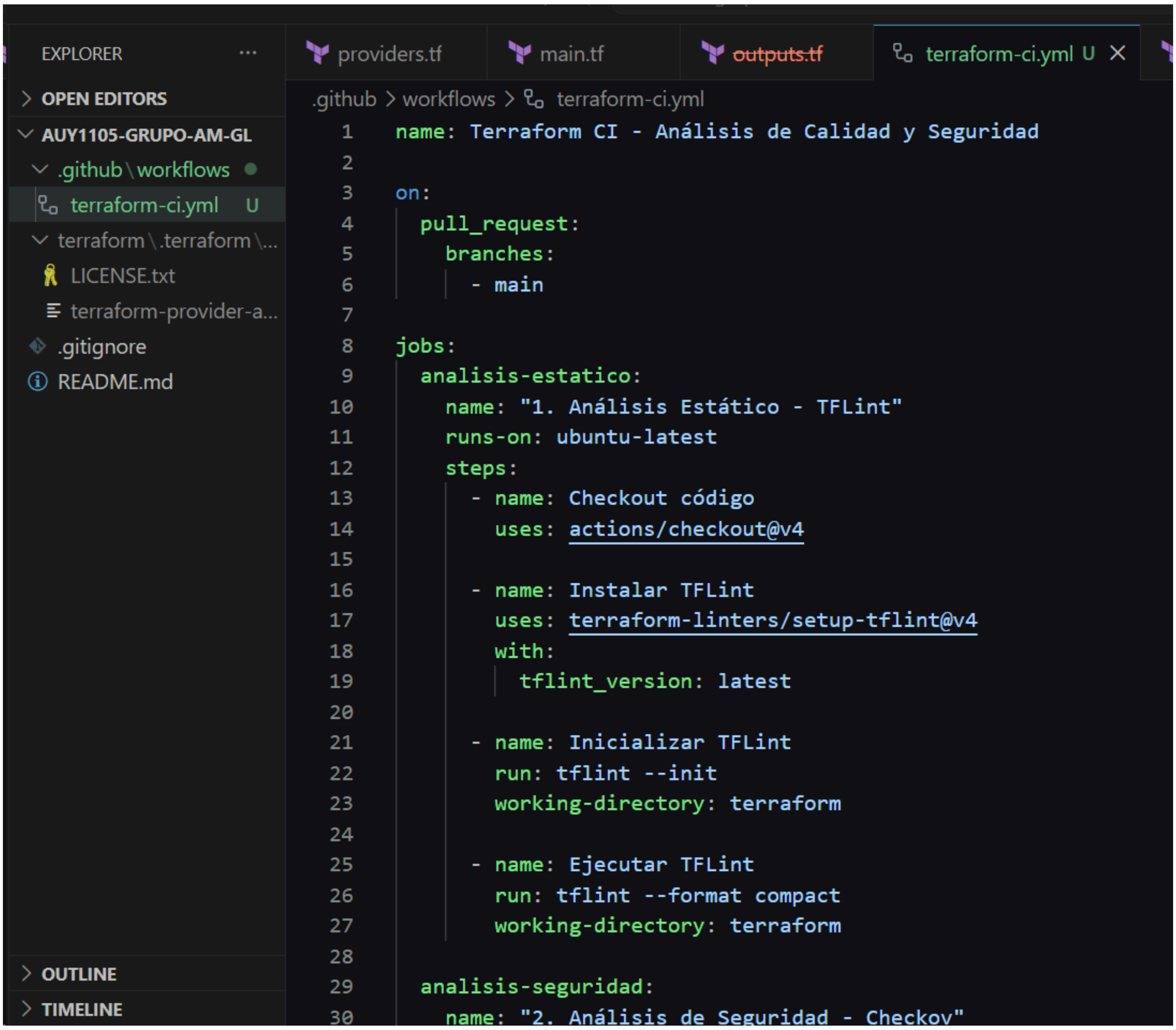
terraform > outputs.tf > output "ec2_instance_id"

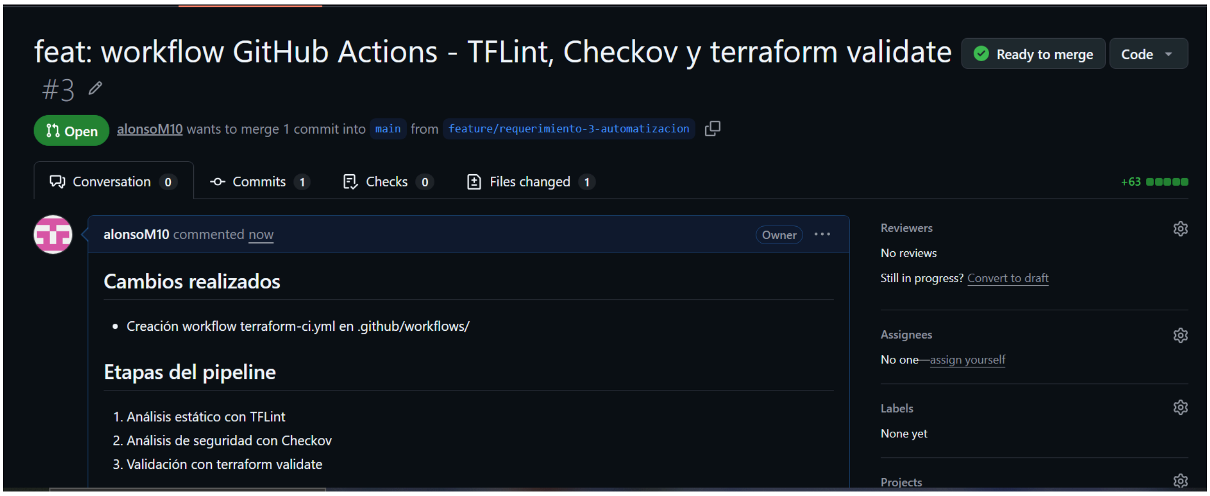
```hcl
output "vpc_id" {
  description = "ID de la VPC"
  value       = aws_vpc.main.id
}

output "subnet_id" {
  description = "ID de la Subnet"
  value       = aws_subnet.public.id
}

output "security_group_id" {
  description = "ID del Security Group"
  value       = aws_security_group.main.id
}

output "ec2_public_ip" {
  description = "IP pública de la EC2"
  value       = aws_instance.main.public_ip
}

```

```
alonso@DESKTOP-VSABIH1 MINGW64 ~/Desktop/Codigo2/AUY1105-grupo-AM-GL (feature/requerimiento-2-terraform)
$ cd ~/Desktop/Codigo2/AUY1105-grupo-AM-GL/terraform
terraform init
Initializing the backend...
Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 5.0"...
- Installing hashicorp/aws v5.100.0...
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

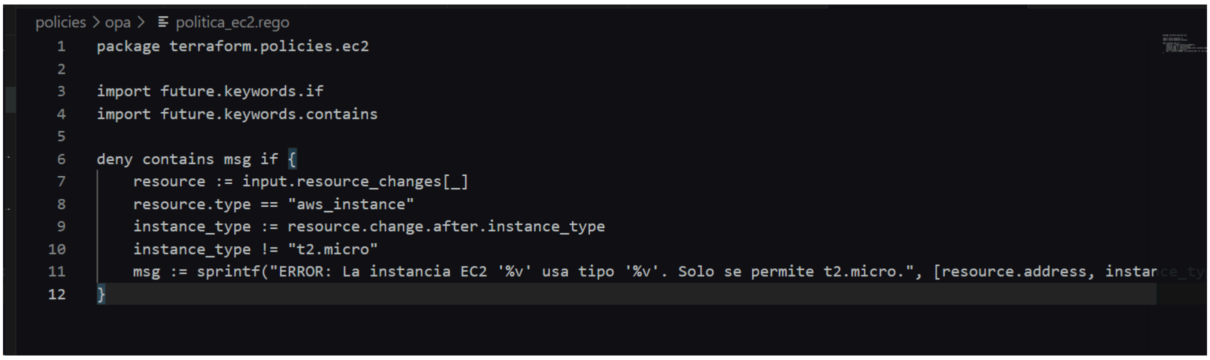
```
alonso@DESKTOP-VSABIH1 MINGW64 ~/Desktop/Codigo2/AUY1105-grupo-AM-GL/terraform (feature/requerimiento-2-terraform)
$ terraform validate
Success! The configuration is valid.
```


feat: código Terraform - VPC, Subnet, Security Group y EC2 #2
Ready to merge
Code
Open
alonsoM10 wants to merge 1 commit into main from feature/requerimiento-2-terraform
Conversation 0
Commits 1
Checks 0
Files changed 5
+146
alonsoM10 commented now
Owner
Cambios realizados
providers.tf: Configuración proveedor AWS v5.0 región us-east-1
variables.tf: Variables del proyecto
main.tf: Definición de VPC, Subnet, Security Group y EC2
outputs.tf: Outputs de los recursos creados
Recursos creados
AUY1105-amgl-vpc (CIDR 10.1.0.0/16)
AUY1105-amgl-subnet (CIDR 10.1.1.0/24)
AUY1105-amgl-sg (Solo SSH entrante)
AUY1105-amgl-ec2 (Ubuntu 24.04 LTS t2.micro)
Reviewers
No reviews
Still in progress? Convert to draft
Assignees
No one—assign yourself
Labels
None yet
Projects
None yet
Milestone
No milestone


EXPLORER
OPEN EDITORS
AUY1105-GRUPO-AM-GL
.github\workflows
terraform-ci.yml U
terraform\.terraform\...
LICENSE.txt
terraform-provider-a...
.gitignore
README.md
OUTLINE
TIMELINE
providers.tf
main.tf
outputs.tf
terraform-ci.yml U
.github > workflows > terraform-ci.yml
name: Terraform CI - Análisis de Calidad y Seguridad
on:
pull_request:
branches:
- main
jobs:
analisis-estatico:
name: "1. Análisis Estático - TFLint"
runs-on: ubuntu-latest
steps:
- name: Checkout código
uses: actions/checkout@v4
- name: Instalar TFLint
uses: terraform-linters/setup-tflint@v4
with:
tflint_version: latest
- name: Inicializar TFLint
run: tflint --init
working-directory: terraform
- name: Ejecutar TFLint
run: tflint --format compact
working-directory: terraform
analisis-seguridad:
name: "2. Análisis de Seguridad - Checkov"


feat: workflow GitHub Actions - TFLint, Checkov y terraform validate
Ready to merge
Code
#3
Open
alonsoM10 wants to merge 1 commit into main from feature/requerimiento-3-automatizacion
Conversation 0
Commits 1
Checks 0
Files changed 1
+63
alonsoM10 commented now
Owner
Cambios realizados
Creación workflow terraform-ci.yml en .github/workflows/
Etapas del pipeline
1. Análisis estático con TFLint
2. Análisis de seguridad con Checkov
3. Validación con terraform validate
Reviewers
No reviews
Still in progress? Convert to draft
Assignees
No one—assign yourself
Labels
None yet
Projects


providers.tf
main.tf
outputs.tf
terraform-ci.yml
politica_ssh.rego U
politica_ec2.rego U
policies > opa > politica_ssh.rego
package terraform.policies.ssh
import future.keywords.if
import future.keywords.contains
deny contains msg if {
resource := input.resource_changes[_]
resource.type == "aws_security_group"
ingress := resource.change.after.ingress[_]
ingress.cidr_blocks[_] == "0.0.0.0/0"
ingress.from_port <= 22
ingress.to_port >= 22
msg := sprintf("ERROR: El Security Group '%v' permite acceso SSH público (0.0.0.0/0). Esto no está permitido.", [
}


policies > opa > politica_ec2.rego
package terraform.policies.ec2
import future.keywords.if
import future.keywords.contains
deny contains msg if {
resource := input.resource_changes[_]
resource.type == "aws_instance"
instance_type := resource.change.after.instance_type
instance_type != "t2.micro"
msg := sprintf("ERROR: La instancia EC2 '%v' usa tipo '%v'. Solo se permite t2.micro.", [resource.address, instan
}